SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name : ELITE SERIES

Product code : 253983

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture : Fragrance mix

1.3 Details of the supplier of the safety data sheet

Company : Symrise AG
Mühlenfeldstrasse 1
37603 Holzminden

Telephone : +495531900

Telefax : +495531901649

E-mail address of person responsible for the SDS : sds@symrise.com

1.4 Emergency telephone number

Emergency CONTACT (24-Hour-Number)
GBK/Infotrac ID 101844: +49(6132)9829021

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Skin sensitisation, Category 1	H317: May cause an allergic skin reaction.
Short-term (acute) aquatic hazard, Category 1	H400: Very toxic to aquatic life.
Long-term (chronic) aquatic hazard, Category 1	H410: Very toxic to aquatic life with long lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Hazard pictograms :

Signal word : Warning

Hazard statements : H317 May cause an allergic skin reaction.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary statements : **Prevention:**
P261 Avoid breathing mist or vapours. P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves.

Response:
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P391 Collect spillage.

Hazardous components which must be listed on the label:
Pentadecan-15-olide
benzyl salicylate
Methyl 2,4-dihydroxy-3,6-dimethylbenzoate
2,2,6-Trimethyl-α-propylcyclohexanepropanol
(Ethoxymethoxy)cyclododecane
(R)-p-mentha-1,8-diene; d-limonene
Linalyl acetate
1H-Indene-ar-propanal, 2,3-dihydro-1,1-dimethyl-
Pin-2(10)-ene
(E)-2-methoxy-4-(prop-1-enyl)phenol
isoeugenol

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Ecological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Toxicological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number		Concentration (% w/w)

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate	259823-90-0 236391-76-7 431-700-8 607-600-00-7 01-0000017792-64-0002	Aquatic Chronic 2; H411	>= 30 - < 50
reaction mass of cis-and trans-cyclohexadec-8-en-1-one	3100-36-5 401-700-2 606-046-00-3 01-0000015154-78-0001, 01-2120921252-67-0000	Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 1	>= 10 - < 20
Pentadecan-15-olide	106-02-5 203-354-6 01-2119987323-31-0000, 01-2119987323-31-0002	Skin Sens. 1B; H317 Aquatic Chronic 2; H411	>= 2,5 - < 10
benzyl salicylate	118-58-1 204-262-9 607-754-00-5	Eye Irrit. 2; H319 Skin Sens. 1B; H317 Aquatic Chronic 3; H412	>= 2,5 - < 10
cis-2-tert-Butylcyclohexyl acetate	20298-69-5 88-41-5 243-718-1	Aquatic Chronic 2; H411	>= 2,5 - < 10
Methyl 2,4-dihydroxy-3,6-dimethylbenzoate	4707-47-5 225-193-0 01-2120762759-36 01-2120762759-36 01-2120762759-36 01-2120762759-36 01-2120762759-36-0007	Skin Sens. 1B; H317	>= 1 - < 10
2,6-Dimethyloct-7-en-2-ol	18479-58-8 242-362-4 01-2119457274-37-	Skin Irrit. 2; H315 Eye Irrit. 2; H319 STOT SE 3; H336 (Central nervous system)	>= 1 - < 10

	0003		
3-Methylcyclopentadecan-1-one	541-91-3 208-795-8 01-2120227688-45	Aquatic Chronic 1; H410 ——— M-Factor (Chronic aquatic toxicity): 1	>= 1 - < 2,5
Oxacycloheptadec-10-en-2-one	28645-51-4 249-120-7 01-2120103324-74	Aquatic Acute 1; H400 Aquatic Chronic 1; H410 ——— M-Factor (Acute aquatic toxicity): 10 M-Factor (Chronic aquatic toxicity): 10	>= 1 - < 2,5
2,2,6-Trimethyl-α-propylcyclohexanepropanol	70788-30-6 274-892-7 01-2120768938-30-0000	Skin Sens. 1B; H317 Aquatic Acute 1; H400 Aquatic Chronic 1; H410 ——— M-Factor (Acute aquatic toxicity): 1 M-Factor (Chronic aquatic toxicity): 1	>= 0,25 - < 1
(Ethoxymethoxy)cyclododecane	58567-11-6 261-332-1 01-2119971571-34-0000, 01-2119971571-34-0002	Skin Irrit. 2; H315 Skin Sens. 1B; H317 Aquatic Chronic 2; H411	>= 0,25 - < 1
(R)-p-mentha-1,8-diene; d-limonene	5989-27-5 227-813-5 601-096-00-2	Flam. Liq. 3; H226 Skin Irrit. 2; H315 Skin Sens. 1B; H317 Asp. Tox. 1; H304 Aquatic Acute 1; H400 Aquatic Chronic 3; H412 ——— M-Factor (Acute aquatic toxicity): 1	>= 0,25 - < 1
musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone	81-14-1 201-328-9 609-069-00-7	Carc. 2; H351 Aquatic Acute 1; H400 Aquatic Chronic 1; H410 ———	>= 0,25 - < 1

		M-Factor (Acute aquatic toxicity): 1 M-Factor (Chronic aquatic toxicity): 1	
Linalyl acetate	115-95-7 204-116-4	Skin Irrit. 2; H315 Eye Irrit. 2; H319 Skin Sens. 1B; H317	>= 0,1 - < 1
1H-Indene-ar-propanal, 2,3-dihydro-1,1-dimethyl-	300371-33-9	Acute Tox. 4; H302 Skin Sens. 1B; H317 Aquatic Chronic 2; H411 --- Acute toxicity estimate Acute oral toxicity: 500 mg/kg	>= 0,1 - < 0,25
Pin-2(10)-ene	127-91-3 204-872-5	Flam. Liq. 3; H226 Skin Irrit. 2; H315 Skin Sens. 1B; H317 Asp. Tox. 1; H304 Aquatic Acute 1; H400 Aquatic Chronic 1; H410	>= 0,1 - < 0,25
p-Mentha-1,4-diene	99-85-4 202-794-6	Flam. Liq. 3; H226 Repr. 2; H361 Aquatic Chronic 2; H411	>= 0,1 - < 0,25
(E)-2-methoxy-4-(prop-1-enyl)phenol	5932-68-3 227-678-2 604-094-00-X	Acute Tox. 4; H302 Acute Tox. 4; H332 Acute Tox. 4; H312 Skin Irrit. 2; H315 Eye Irrit. 2; H319 Skin Sens. 1A; H317 STOT SE 3; H335 (Respiratory system) --- specific concentration limit Skin Sens. 1A; H317 >= 0,01 % --- Acute toxicity estimate Acute oral toxicity: 1.500 mg/kg	< 0,01

		Acute dermal toxicity: 1.912 mg/kg	
isoeugenol	97-54-1 202-590-7 604-094-00-X	Acute Tox. 4; H302 Acute Tox. 4; H332 Acute Tox. 4; H312 Skin Irrit. 2; H315 Eye Irrit. 2; H319 Skin Sens. 1A; H317 STOT SE 3; H335 (Respiratory system) specific concentration limit Skin Sens. 1A; H317 >= 0,01 % Acute toxicity estimate Acute oral toxicity: 1.560 mg/kg	< 0,01

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

General advice	:	Move out of dangerous area. Show this safety data sheet to the doctor in attendance. Do not leave the victim unattended.
If inhaled	:	If unconscious, place in recovery position and seek medical advice. Keep respiratory tract clear. If symptoms persist, call a physician.
In case of skin contact	:	If skin irritation persists, call a physician. If on skin, rinse well with water. If on clothes, remove clothes.
In case of eye contact	:	Flush eyes with water as a precaution. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.
If swallowed	:	Keep respiratory tract clear. Never give anything by mouth to an unconscious person. If symptoms persist, call a physician.

4.2 Most important symptoms and effects, both acute and delayed

Risks : May cause an allergic skin reaction.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : Treat symptomatically.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Unsuitable extinguishing media : High volume water jet

5.2 Special hazards arising from the substance or mixture

Hazardous combustion products : No hazardous combustion products are known

5.3 Advice for firefighters

Special protective equipment for firefighters : In the event of fire, wear self-contained breathing apparatus.

Further information : Standard procedure for chemical fires.
Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Use personal protective equipment.
Ensure adequate ventilation.
Evacuate personnel to safe areas.

6.2 Environmental precautions

Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust).
Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

See sections: 7, 8, 11, 12 and 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.

Advice on protection against fire and explosion : Normal measures for preventive fire protection.

Hygiene measures : General industrial hygiene practice. When using do not eat or drink. When using do not smoke. Wash hands before breaks and at the end of workday.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : No smoking. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Electrical installations / working materials must comply with the technological safety standards.

Advice on common storage : No special restrictions on storage with other products.

Storage class (TRGS 510) : 10

Further information on storage stability : No decomposition if stored and applied as directed.

7.3 Specific end use(s)

Specific use(s) : Fragrance mix

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Components	CAS-No.	Value type (Form of exposure)	Control parameters	Basis
Oxydipropanol	25265-71-8	MAK (Vapor and aerosol, inhalable fraction.)	100 mg/m3	DFG

		(Vapor and aerosol, inhalable fraction.)	100 mg/m3	DE TRGS 900
	Further information: Sum of vapour and aerosols			
(R)-p-mentha-1,8-diene; d-limonene	5989-27-5	MAK	5 ppm 28 mg/m3	DFG
			5 ppm 28 mg/m3	DE TRGS 900
ethanol; ethyl alcohol	64-17-5	MAK	200 ppm 380 mg/m3	DFG
			200 ppm 380 mg/m3	DE TRGS 900

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006

Substance name	End Use	Exposure routes	Potential health effects	Value
2,6-Dimethyloct-7-en-2-ol	Workers	Inhalation	Long-term systemic effects	24,7 mg/m3
	Workers	Skin contact	Long-term systemic effects	7 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	4,35 mg/m3
	Consumers	Skin contact	Long-term systemic effects	2,5 mg/kg bw/day
	Consumers	Ingestion	Long-term systemic effects	2,5 mg/kg bw/day
tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans)	Workers	Inhalation	Long-term systemic effects	44,1 mg/m3
	Workers	Skin contact	Long-term systemic effects	41,7 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	13 mg/m3
	Consumers	Skin contact	Long-term systemic effects	25 mg/kg bw/day
	Consumers	Ingestion	Long-term systemic effects	7,5 mg/kg bw/day
2,2,6-Trimethyl-α-propylcyclohexanepropanol	Workers	Inhalation	Long-term systemic effects	9,87 mg/m3
	Workers	Skin contact	Long-term systemic effects	2,8 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	1,74 mg/m3
	Consumers	Skin contact	Long-term systemic effects	1 mg/kg bw/day
	Consumers	Ingestion	Long-term systemic effects	0,5 mg/kg bw/day
(Ethoxymeth-	Workers	Inhalation	Long-term systemic	23,5 mg/m3

oxy)cyclododecane			effects	
	Workers	Skin contact	Long-term systemic effects	3,3 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	5,8 mg/m3
	Consumers	Skin contact	Long-term systemic effects	1,67 mg/kg bw/day
	Consumers	Ingestion	Long-term systemic effects	1,67 mg/kg bw/day
reaction mass of cis- and trans-cyclohexadec-8-en-1-one	Workers	Inhalation	Long-term systemic effects	3,822 mg/m3
	Workers	Skin contact	Long-term systemic effects	1,084 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	0,674 mg/m3
	Consumers	Skin contact	Long-term systemic effects	0,387 mg/kg bw/day
	Consumers	Ingestion	Long-term systemic effects	0,387 mg/kg bw/day
(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohex-yl)ethoxycarbonyl]methyl propanoate	Workers	Inhalation	Long-term systemic effects	32,44 mg/m3
	Workers	Skin contact	Long-term systemic effects	9,2 mg/kg bw/day
	Consumers	Inhalation	Long-term systemic effects	4,87 mg/m3
	Consumers	Skin contact	Long-term systemic effects	3,29 mg/kg bw/day
	Consumers	Ingestion	Long-term systemic effects	3,29 mg/kg bw/day

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006

Substance name	Environmental Compartment	Value
2,6-Dimethyloct-7-en-2-ol	Fresh water	0,0278 mg/l
	Fresh water sediment	0,594 mg/kg dry weight (d.w.)
	Marine water	0,00278 mg/l
	Marine sediment	0,059 mg/kg dry weight (d.w.)
	Sewage treatment plant	10 mg/l
	Soil	0,103 mg/kg dry weight (d.w.)
tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans)	Fresh water	0,094 mg/l
	Fresh water sediment	0,412 mg/kg dry weight (d.w.)
	Marine water	0,0094 mg/l

	Marine sediment	0,0412 mg/kg dry weight (d.w.)
	Sewage treatment plant	10 mg/l
	Soil	0,09 mg/kg dry weight (d.w.)
2,2,6-Trimethyl-α-propylcyclohexanepropanol	Fresh water	0,000696 mg/l
	Fresh water sediment	0,5 mg/kg dry weight (d.w.)
	Marine water	0,000070 mg/l
	Marine sediment	0,05 mg/kg dry weight (d.w.)
	Sewage treatment plant	100 mg/l
	Soil	0,396 mg/kg dry weight (d.w.)
(Ethoxymethoxy)cyclododecane	Fresh water	0,002 mg/l
	Fresh water sediment	2,35 mg/kg dry weight (d.w.)
	Marine water	0,00016 mg/l
	Marine sediment	0,235 mg/kg dry weight (d.w.)
	Sewage treatment plant	100 mg/l
	Soil	0,468 mg/kg dry weight (d.w.)
reaction mass of cis-and trans-cyclohexadec-8-en-1-one	Fresh water	0,00028 mg/l
	Fresh water sediment	0,3391 mg/kg dry weight (d.w.)
	Marine water	0,000028 mg/l
	Marine sediment	0,03391 mg/kg dry weight (d.w.)
	Sewage treatment plant	100 mg/l
	Soil	3,7 mg/kg dry weight (d.w.)
(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohex-yl)ethoxycarbonyl]methyl propanoate	Fresh water	0,014 mg/l
	Fresh water sediment	15,45 mg/kg dry weight (d.w.)
	Marine water	0,0014 mg/l
	Marine sediment	1,55 mg/kg dry weight (d.w.)
	Sewage treatment plant	10 mg/l
	Soil	3,08 mg/kg dry weight (d.w.)
Pentadecan-15-olide	Fresh water	0,0027 mg/l
	Fresh water sediment	21 mg/kg dry weight (d.w.)
	Marine water	0,00027 mg/l
	Marine sediment	4,2 mg/kg dry weight (d.w.)

	Sewage treatment plant	10 mg/l
	Soil	5,44 mg/kg dry weight (d.w.)

8.2 Exposure controls

Personal protective equipment

Eye/face protection : Eye wash bottle with pure water
Tightly fitting safety goggles

Hand protection

Remarks : Take note of the information given by the producer concerning permeability and break through times, and of special workplace conditions (mechanical strain, duration of contact). As the product is a mixture of several substances, the durability of the glove materials cannot be calculated in advance and has to be tested before use. Wear chemicals-resistant gloves, e.g. safety gloves of nitril (thickness 0.4mm) or of butyl rubber (thickness 0.7mm). The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Respiratory protection : Not required; except in case of aerosol formation.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state : clear liquid

Colour : colorless to yellow

Odour : characteristic

Odour Threshold : No data available

Melting point/freezing point : not determined

Boiling point/boiling range : not determined

Flammability : not determined

Upper explosion limit / Upper flammability limit : Vapours may form explosive mixtures with air.

Lower explosion limit / Lower flammability limit : Vapours may form explosive mixtures with air.

Flash point : > 100 °C

Decomposition temperature : not determined

pH : Not applicable

Viscosity
- Viscosity, dynamic : not determined
- Viscosity, kinematic : not determined

Solubility(ies)
- Water solubility : immiscible

Partition coefficient: n-octanol/water : Not applicable

Vapour pressure : < 1 kPa (50 °C)
calculated

Relative density : not determined

Bulk density : Not applicable

Relative vapour density : not determined

9.2 Other information

Explosives : Due to its structural properties, the product is not classified as explosive

Oxidizing properties : The substance or mixture is not classified as oxidizing.

Self-ignition : The substance or mixture is not classified as self heating.

Evaporation rate : Not applicable

Molecular weight : Not applicable

SECTION 10: Stability and reactivity

10.1 Reactivity

No decomposition if stored and applied as directed.

10.2 Chemical stability

No decomposition if stored and applied as directed.

10.3 Possibility of hazardous reactions

Hazardous reactions : No decomposition if stored and applied as directed.

10.4 Conditions to avoid

Conditions to avoid : No data available

10.5 Incompatible materials

Materials to avoid : Not applicable

10.6 Hazardous decomposition products

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Not classified based on available information.

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Acute oral toxicity : LD50 Oral (Rat, male and female): > 2.000 mg/kg
Method: OECD 423
GLP: yes

Acute dermal toxicity : LD50 Dermal (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Acute oral toxicity : LD50 (Rat, male and female): > 10.000 mg/kg
Method: OECD Test Guideline 401
GLP: yes

Acute dermal toxicity : LD50 (Rabbit, male and female): > 4.600 mg/kg
GLP: yes

benzyl salicylate:

Acute oral toxicity : LD50 Oral (Rat): 2.227 mg/kg

cis-2-tert-Butylcyclohexyl acetate:

Acute oral toxicity : LD50 Oral (Rat): 4.600 mg/kg
Method: OECD Test Guideline 401
GLP: no

Acute dermal toxicity : LD50 Dermal (Rabbit): > 5.000 mg/kg
Method: OECD Test Guideline 402
GLP: no

Methyl 2,4-dihydroxy-3,6-dimethylbenzoate:

Acute oral toxicity : LD50 Oral (Rat, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 401
GLP: no

Acute dermal toxicity : LD50 Dermal (Rat, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 402
GLP: no

2,6-Dimethyloct-7-en-2-ol:

Acute oral toxicity : LD50 Oral (Rat): 3.600 mg/kg

3-Methylcyclopentadecan-1-one:

Acute oral toxicity : LD50 (Rat): > 5.000 mg/kg
Method: OECD Test Guideline 401
Remarks: Weight of Evidence

Acute dermal toxicity : LD50 (Rabbit): > 5.000 mg/kg
Method: OECD Test Guideline 402

Oxacycloheptadec-10-en-2-one:

Acute oral toxicity : (Rat, female): > 2.000 mg/kg
Method: OECD 423
GLP: yes

Acute dermal toxicity : LD50 (Rat, female): > 2.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Acute oral toxicity : LD50 (Rat, male and female): > 18.000 mg/kg
Method: OECD Test Guideline 401
GLP: no

Acute dermal toxicity : LD50 (Rabbit, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes

(Ethoxymethoxy)cyclododecane:

Acute oral toxicity : LD50 Oral (Rat, male): > 5.000 mg/kg
Method: OECD Test Guideline 401
GLP: yes

Acute dermal toxicity : LD50 Dermal (Rabbit, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes

(R)-p-mentha-1,8-diene; d-limonene:

Acute oral toxicity : LD50 Oral (Rat, female): > 2.000 mg/kg
Method: OECD Test Guideline 423
GLP: yes

Acute dermal toxicity : LD50 (Rabbit): > 5.000 mg/kg
Remarks: Information given is based on data obtained from similar substances.

musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone:

Acute oral toxicity : LD50 (Rat, male and female): > 5.000 mg/kg
Method: OECD Test Guideline 401
GLP: No information available.

Acute inhalation toxicity : LC50 (Rat, male and female): > 2,99 mg/l
Exposure time: 4 h
Test atmosphere: dust/mist
Method: OECD Test Guideline 403
GLP: yes

Acute dermal toxicity : LD50 (Rabbit): > 10.000 mg/kg

Linalyl acetate:

Acute oral toxicity : LD50 (Rat, male and female): > 9.000 mg/kg
GLP: no

Acute dermal toxicity : LD50 (Rabbit): > 5.000 mg/kg
GLP: no

Acute toxicity (other routes of administration) : LD50 (Mouse): 721 mg/kg
Method: Intraperitoneal toxicity

1H-Indene-ar-propanal, 2,3-dihydro-1,1-dimethyl-:

Acute oral toxicity : Acute toxicity estimate: 500 mg/kg

Pin-2(10)-ene:

Acute oral toxicity : LD50 (Rat): > 5.000 mg/kg

Acute dermal toxicity : LD50 (Rabbit): > 5.000 mg/kg

p-Mentha-1,4-diene:

Acute oral toxicity : LD50 Oral (Rat, female): > 2.000 mg/kg
Method: OECD 423
GLP: yes

Acute dermal toxicity : LD50 Dermal (Rat, male and female): > 2.000 mg/kg
Method: OECD Test Guideline 402
GLP: yes

(E)-2-methoxy-4-(prop-1-enyl)phenol:

Acute oral toxicity : LD50 Oral (Rat): 1.500 mg/kg

Acute dermal toxicity	:	LD50 Dermal (Rabbit): 1.912 mg/kg Method: OECD Test Guideline 402 GLP: yes

isoeugenol:

Acute oral toxicity	:	LD50 (Rat): 1.560 mg/kg

Skin corrosion/irritation

Not classified based on available information.

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Species	:	Rabbit
Exposure time	:	4 h
Method	:	OECD Test Guideline 404
Result	:	No skin irritation
GLP	:	yes
Dose	:	0,5 ml
Concentration	:	100 %

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Species	:	Rabbit
Exposure time	:	24 h
Method	:	OECD Test Guideline 404
Result	:	No skin irritation
GLP	:	yes
Dose	:	500 mg
Concentration	:	100 %

Pentadecan-15-olide:

Species	:	Rabbit
Exposure time	:	4 h
Method	:	OECD Test Guideline 404
Result	:	Mild skin irritation
GLP	:	yes
Dose	:	0,5 ml
Concentration	:	100 %

benzyl salicylate:

Species	:	Humans
Result	:	No skin irritation
Concentration	:	30 %

Species	:	Rabbit
Exposure time	:	4 h
Method	:	OECD Test Guideline 404
Result	:	No skin irritation
GLP	:	yes
Concentration	:	100 %

cis-2-tert-Butylcyclohexyl acetate:

Species	:	Rabbit
Exposure time	:	4 h
Method	:	OECD Test Guideline 404
Result	:	Mild skin irritation
GLP	:	no
Concentration	:	100 %

Methyl 2,4-dihydroxy-3,6-dimethylbenzoate:

Species	:	reconstructed human epidermis (RhE)
Method	:	OECD 439
Result	:	No skin irritation
GLP	:	yes
Dose	:	0,025 ml
Concentration	:	100 %

3-Methylcyclopentadecan-1-one:

Species	:	reconstructed human epidermis (RhE)
Method	:	OECD 439
Result	:	No skin irritation
GLP	:	yes
Concentration	:	100 %

Oxacycloheptadec-10-en-2-one:

Species	:	Humans
Result	:	No skin irritation
Concentration	:	6 %

Species	:	Rat
Exposure time	:	24 h
Result	:	No skin irritation
GLP	:	yes
Concentration	:	100 %

Species	:	reconstructed human epidermis (RhE)
Method	:	OECD 439
Result	:	No skin irritation
GLP	:	yes
Concentration	:	100 %

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Species	:	reconstructed human epidermis (RhE)
Exposure time	:	1 h
Method	:	OECD Test Guideline 439
Result	:	No skin irritation
GLP	:	yes
Concentration	:	100 %

(Ethoxymethoxy)cyclododecane:

Species	:	Rabbit
Exposure time	:	4 h
Method	:	OECD Test Guideline 404
Result	:	Irritating to skin.
GLP	:	yes
Dose	:	0,5 ml
Concentration	:	100 %

(R)-p-mentha-1,8-diene; d-limonene:

Species	:	Rabbit
Exposure time	:	4 h
Method	:	OECD Test Guideline 404
Result	:	Skin irritation
GLP	:	yes
Concentration	:	100 %

musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone:

Species	:	reconstructed human epidermis (RhE)
Exposure time	:	1 h
Method	:	OECD Test Guideline 439
Result	:	No skin irritation
GLP	:	yes
Dose	:	25 mg
Concentration	:	100 %

Linalyl acetate:

Species	:	Rabbit
Exposure time	:	4 h
Method	:	OECD Test Guideline 404
Result	:	Skin irritation
GLP	:	No information available.
Concentration	:	100 %

Pin-2(10)-ene:

Species	:	Rabbit
Exposure time	:	24 h
Result	:	Moderate irritation of skin

Species	:	Humans
Exposure time	:	48 h
Method	:	Closed patch test
Result	:	No skin irritation
Concentration	:	12 %
solvents	:	Petrolatum

p-Mentha-1,4-diene:

Species	:	reconstructed human epidermis (RhE)

Exposure time	:	1 h
Method	:	OECD Test Guideline 439
Result	:	No skin irritation
GLP	:	yes
Concentration	:	100 %

Serious eye damage/eye irritation

Not classified based on available information.

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Species	:	Rabbit
Method	:	OECD Test Guideline 405
Result	:	No eye irritation
GLP	:	yes
Dose	:	0,1 ML
Concentration	:	100 %

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Species	:	Rabbit
Method	:	OECD Test Guideline 405
Result	:	No eye irritation
GLP	:	yes
Dose	:	0,1 ML
Concentration	:	10 %

Pentadecan-15-olide:

Species	:	Rabbit
Method	:	OECD Test Guideline 405
Result	:	Mild eye irritation
GLP	:	yes
Dose	:	0,1 ML
Concentration	:	100 %
Remarks	:	Information given is based on data obtained from similar substances.

benzyl salicylate:

Species	:	Rabbit
Method	:	Draize Test
Result	:	Eye irritation
GLP	:	no
Dose	:	0,1 ML

cis-2-tert-Butylcyclohexyl acetate:

Species	:	Rabbit
Exposure time	:	24 h
Method	:	OECD Test Guideline 405
Result	:	No eye irritation
GLP	:	no

Dose : 0,1 ML
Concentration : 100 %

Methyl 2,4-dihydroxy-3,6-dimethylbenzoate:

Species : Rabbit
Method : OECD Test Guideline 405
Result : Mild eye irritation
GLP : no
Concentration : 100 %

2,6-Dimethyloct-7-en-2-ol:

Species : Rabbit
Result : Eye irritation
GLP : yes
Dose : 0,1 ML
Concentration : 100 %

3-Methylcyclopentadecan-1-one:

Species : Rabbit
Method : OECD Test Guideline 405
Result : No eye irritation
GLP : yes
Dose : 0,1 ML
Concentration : 100 %

Oxacycloheptadec-10-en-2-one:

Species : Rabbit
Method : Draize Test
Result : Mild eye irritation
GLP : No information available.
Dose : 0,1 ML
Concentration : 0,5 %

Species : Human EpiOcular Eye Model Test
Method : OECD Test Guideline 492
Result : No eye irritation
GLP : yes
Concentration : 100 %

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Species : Bovine cornea
Method : OECD Test Guideline 437
Result : No eye irritation
GLP : yes
Dose : 0,75 ML
Concentration : 100 %

(Ethoxymethoxy)cyclododecane:

Species	:	Rabbit
Method	:	OECD Test Guideline 405
Result	:	Mild eye irritation
GLP	:	yes
Dose	:	0,1 ML
Concentration	:	100 %

(R)-p-mentha-1,8-diene; d-limonene:

Species	:	Rabbit
Method	:	OECD Test Guideline 405
Result	:	Mild eye irritation
GLP	:	yes
Concentration	:	100 %

musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone:

Species	:	Rabbit
Method	:	OECD Test Guideline 405
Result	:	No eye irritation
GLP	:	yes
Dose	:	70 MG
Concentration	:	100 %

Linalyl acetate:

Species	:	Rabbit
Result	:	Eye irritation
GLP	:	no
Concentration	:	100 %

p-Mentha-1,4-diene:

Species	:	Bovine cornea
Method	:	OECD 437
Result	:	Mild eye irritation
GLP	:	yes
Concentration	:	100 %

Respiratory or skin sensitisation

Skin sensitisation

May cause an allergic skin reaction.

Respiratory sensitisation

Not classified based on available information.

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Test Type	:	Maximisation Test
Species	:	Guinea pig

Method	:	OECD Test Guideline 406
Result	:	No sensitizing effect.
GLP	:	yes
Concentration	:	75 %
solvents	:	Peanut oil

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Test Type	:	Magnusson & Kligmann test
Species	:	Guinea pig
Result	:	No sensitizing effect.
GLP	:	yes
Concentration	:	25 %
solvents	:	Vaseline

Test Type	:	HRIPT
Species	:	Humans
Result	:	No sensitizing effect.
GLP	:	no
Concentration	:	20 %
solvents	:	Diethylphthalate/Ethyl alcohol (1:1)

Pentadecan-15-olide:

Test Type	:	HRIPT
Species	:	Humans
Result	:	No sensitizing effect.
Concentration	:	10 %
solvents	:	Diethylphthalate/Ethyl alcohol (3:1)

Test Type	:	Local Lymph Node Assay
Species	:	Mouse
Method	:	OECD 429
Result	:	Sensitizing effect.
GLP	:	yes
Concentration	:	25,5 %
solvents	:	Acetone/Olive oil (4:1)

benzyl salicylate:

Test Type	:	Local Lymph Node Assay
Species	:	Mouse
Method	:	OECD 429
Result	:	Sensitizing effect.
GLP	:	yes
Concentration	:	2,9 %
solvents	:	Diethylphthalate/Ethyl alcohol (3:1)

cis-2-tert-Butylcyclohexyl acetate:

Test Type	:	Buehler Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	No sensitizing effect.

GLP : no
Concentration : 100 %

Methyl 2,4-dihydroxy-3,6-dimethylbenzoate:

Test Type : Local Lymph Node Assay
Species : Mouse
Method : OECD 429
Result : Sensitizing effect.
GLP : yes
Concentration : 19 %
solvents : Diethylphthalate/Ethyl alcohol (3:1)

2,6-Dimethyloct-7-en-2-ol:

Species : Humans
Result : No sensitizing effect.
Rate of positive effects : 0/25
Concentration : 4 %

Test Type : Maximisation Test
Species : Guinea pig
Method : OECD Test Guideline 406
Result : No sensitizing effect.
GLP : yes
Concentration : 100 %

3-Methylcyclopentadecan-1-one:

Species : Humans
Result : No sensitizing effect.
Concentration : 30 %

Oxacycloheptadec-10-en-2-one:

Species : Humans
Result : No sensitizing effect.
GLP : No information available.
Rate of positive effects : 0/54
Concentration : 6 %

Species : Humans
Result : No sensitizing effect.
GLP : No information available.
Rate of positive effects : 0/422
Concentration : 5 %
solvents : Petrolatum

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Test Type : Local Lymph Node Assay
Species : Mouse
Method : OECD Test Guideline 442B
Result : Sensitizing effect.

GLP	:	yes
Concentration	:	12,73 %
solvents	:	Acetone/Olive oil (4:1)

(Ethoxymethoxy)cyclododecane:

Test Type	:	Local Lymph Node Assay
Species	:	Mouse
Method	:	OECD 429
Result	:	Sensitizing effect.
GLP	:	yes
Concentration	:	25,1 %
solvents	:	Acetone/Olive oil (4:1)

(R)-p-mentha-1,8-diene; d-limonene:

Test Type	:	Local Lymph Node Assay
Species	:	Mouse
Method	:	OECD Test Guideline 429
Result	:	Sensitizing effect.
GLP	:	yes
Concentration	:	22 %
solvents	:	Diethylphthalate/Ethyl alcohol (3:1)

musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone:

Test Type	:	Maximisation Test
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	No sensitizing effect.
GLP	:	yes
Rate of positive effects	:	0/20
solvents	:	Acetone/Olive oil (4:1)

Pin-2(10)-ene:

Test Type	:	Maximisation Test
Species	:	Humans
Result	:	No sensitizing effect.
Concentration	:	12 %
solvents	:	Petrolatum

p-Mentha-1,4-diene:

Species	:	Humans
Result	:	No sensitizing effect.
Concentration	:	5 %

(E)-2-methoxy-4-(prop-1-enyl)phenol:

Test Type	:	Local Lymph Node Assay
Species	:	Mouse
Method	:	OECD 429

Result	:	Sensitizing effect.
GLP	:	yes
Concentration	:	1,54 %
solvents	:	Acetone/Olive oil (4:1)

Germ cell mutagenicity

Not classified based on available information.

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Genotoxicity in vitro	:	Test Type: Ames test Metabolic activation: with and without metabolic activation Method: OECD 471 Result: negative GLP: yes Test Type: In vitro Mammalian Chromosome Aberration Test Test system: Human lymphocytes Metabolic activation: with and without metabolic activation Method: OECD 473 Result: negative GLP: yes Test Type: In vitro Mammalian Cell Gene Mutation Test Test system: Chinese hamster cells Metabolic activation: with and without metabolic activation Method: OECD 476 Result: negative GLP: yes

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Genotoxicity in vitro	:	Test Type: Ames test Metabolic activation: with and without metabolic activation Method: OECD 471 Result: negative GLP: yes Test Type: In vitro Mammalian Cell Gene Mutation Test Test system: V79 cells Method: OECD 476 Result: negative GLP: yes
Genotoxicity in vivo	:	Test Type: Mammalian Erythrocyte Micronucleus Test Species: Mouse (male and female) Cell type: Bone marrow Application Route: Oral Method: OECD 474 Result: negative GLP: yes

Pentadecan-15-olide:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative
GLP: yes

benzyl salicylate:

Genotoxicity in vitro : Test Type: Ames test
Test system: Mammal cells
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative

cis-2-tert-Butylcyclohexyl acetate:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative
GLP: yes

Test Type: In vitro Mammalian Chromosome Aberration Test
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Method: OECD 473
Result: negative
GLP: yes

Test Type: In vitro Mammalian Cell Gene Mutation Test
Test system: mouse lymphoma L5178Y cells
Metabolic activation: with and without metabolic activation
Method: OECD 476
Result: negative
GLP: yes

Methyl 2,4-dihydroxy-3,6-dimethylbenzoate:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative
GLP: yes

Test Type: In vitro Mammalian Chromosome Aberration Test
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Method: OECD 473
Result: positive
GLP: yes

Genotoxicity in vivo : Test Type: Mammalian Erythrocyte Micronucleus Test
Species: Mouse (male and female)

Application Route: Intraperitoneal injection
Method: OECD Test Guideline 474
Result: negative
GLP: yes

2,6-Dimethyloct-7-en-2-ol:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative
GLP: yes

Test Type: In vitro Mammalian Cell Gene Mutation Test
Test system: mouse lymphoma L5178Y cells
Metabolic activation: with and without metabolic activation
Method: OECD 476
Result: negative
GLP: yes

Test Type: In vitro Mammalian Chromosome Aberration Test
Test system: Human lymphocytes
Metabolic activation: with and without metabolic activation
Method: OECD 473
Result: negative
GLP: yes

3-Methylcyclopentadecan-1-one:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative
GLP: yes

Oxacycloheptadec-10-en-2-one:

Genotoxicity in vitro : Test Type: Ames test
Test system: TA98
Metabolic activation: with and without metabolic activation
Method: OECD 471

Test Type: Ames test
Test system: TA100
Metabolic activation: with and without metabolic activation
Method: OECD 471

Test Type: Ames test
Test system: TA102
Metabolic activation: with and without metabolic activation
Method: OECD 471

Test Type: Ames test
Test system: TA1535

Metabolic activation: with and without metabolic activation
Method: OECD 471

Test Type: Ames test
Test system: TA1537
Metabolic activation: with and without metabolic activation
Method: OECD 471

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative
GLP: yes

Test Type: In Vitro Mammalian Cell Micronucleus Test
Test system: Human lymphocytes
Metabolic activation: with and without metabolic activation
Method: OECD 487
Result: negative
GLP: yes

Test Type: In vitro Mammalian Cell Gene Mutation Test
Test system: V79 cells
Metabolic activation: with and without metabolic activation
Method: OECD 476
Result: negative
GLP: yes

(Ethoxymethoxy)cyclododecane:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative
GLP: yes

Test Type: In Vitro Mammalian Cell Micronucleus Test
Test system: Human lymphocytes
Metabolic activation: with and without metabolic activation
Method: OECD 487
Result: negative
GLP: yes

Test Type: In vitro Mammalian Cell Gene Mutation Test
Test system: V79 cells
Metabolic activation: with and without metabolic activation
Method: OECD 476
Result: negative
GLP: yes

Test Type: In vitro Mammalian Chromosome Aberration Test
Test system: Human lymphocytes

Metabolic activation: with and without metabolic activation
Method: OECD 473
Result: negative
GLP: yes

(R)-p-mentha-1,8-diene; d-limonene:

Genotoxicity in vitro : Test Type: In vitro Mammalian Cell Gene Mutation Test
Test system: mouse lymphoma L5178Y cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative

Test Type: sister chromatid exchange assay
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 479
Result: negative
GLP: no

Test Type: In vitro Mammalian Chromosome Aberration Test
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: negative
GLP: no

Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes

Genotoxicity in vivo : Test Type: comet assay
Species: Rat (male)
Application Route: Oral
Result: negative
GLP: no

musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative
GLP: yes

Test Type: In vitro Mammalian Chromosome Aberration Test
Test system: Chinese hamster ovary cells
Metabolic activation: with and without metabolic activation
Method: OECD 473
Result: negative
GLP: No information available.

Test Type: In vitro Mammalian Cell Gene Mutation Test
Test system: mouse lymphoma L5178Y cells
Metabolic activation: with and without metabolic activation
Method: OECD 476
Result: negative
GLP: yes

Linalyl acetate:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes

Test Type: In vitro Mammalian Chromosome Aberration Test
Test system: Human lymphocytes
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: negative
GLP: yes

Test Type: In vitro Mammalian Cell Gene Mutation Test
Test system: mouse lymphoma L5178Y cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative
GLP: yes

Genotoxicity in vivo : Test Type: Mammalian Erythrocyte Micronucleus Test
Species: Mouse (male and female)
Strain: CD1
Cell type: Bone marrow
Application Route: Oral
Method: OECD Test Guideline 474
Result: negative
GLP: yes

p-Mentha-1,4-diene:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD 471
Result: negative
GLP: yes

Test Type: In vitro Mammalian Cell Gene Mutation Test
Test system: mouse lymphoma L5178Y cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 490
Result: negative
GLP: yes

Test Type: In Vitro Mammalian Cell Micronucleus Test
Test system: mouse lymphoma L5178Y cells
Metabolic activation: with and without metabolic activation
Method: OECD 487
Result: negative
GLP: yes

Carcinogenicity

Not classified based on available information.

Reproductive toxicity

Not classified based on available information.

STOT - single exposure

Not classified based on available information.

STOT - repeated exposure

Not classified based on available information.

Repeated dose toxicity

<u>Components:</u>

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Species	:	Rat, male and female
NOAEL	:	917 mg/kg
Application Route	:	Oral
Exposure time	:	28 d
Method	:	OECD Test Guideline 407

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Species	:	Rat, male and female
NOAEL	:	600 mg/kg bw/day
Application Route	:	Oral
Method	:	OECD 422
GLP	:	yes

Linalyl acetate:

Species	:	Rat, male and female
NOAEL	:	160 mg/kg
Application Route	:	Oral
Exposure time	:	28 d
Number of exposures	:	daily
Method	:	OECD Test Guideline 407
GLP	:	yes

Species	:	Rat, male and female
NOAEL	:	250 mg/kg
Application Route	:	Dermal
Exposure time	:	91 d
Number of exposures	:	daily
Method	:	OECD Test Guideline 411

GLP : yes

Aspiration toxicity

Not classified based on available information.

11.2 Information on other hazards

Endocrine disrupting properties

Not classified based on available information.

Product:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Pentadecan-15-olide:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Further information

Product:

Remarks : No data available

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Remarks : No data available

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Remarks : No data available

Pentadecan-15-olide:

Remarks : No data available

Methyl 2,4-dihydroxy-3,6-dimethylbenzoate:

Remarks : No data available

2,6-Dimethyloct-7-en-2-ol:

Remarks : Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.
Concentrations substantially above the TLV value may cause narcotic effects.
Solvents may degrease the skin.

3-Methylcyclopentadecan-1-one:

Remarks : No data available

Oxacycloheptadec-10-en-2-one:

Remarks : No data available

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Remarks : No data available

(Ethoxymethoxy)cyclododecane:

Remarks : No data available

SECTION 12: Ecological information

12.1 Toxicity

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 1,6 mg/l
End point: mortality
Exposure time: 96 h
Test Type: semi-static test
Analytical monitoring: yes
Method: OECD Test Guideline 203
GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 3,16 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: semi-static test
Analytical monitoring: yes
Method: OECD Test Guideline 202
GLP: yes

Toxicity to algae/aquatic plants : EC50 (Desmodesmus subspicatus (green algae)): > 3,73 mg/l
End point: Growth rate
Exposure time: 96 h

Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

NOEC (Desmodesmus subspicatus (green algae)): > 3,73 mg/l
End point: Growth rate
Exposure time: 96 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

Toxicity to microorganisms : NOEC (Activated sludge): > 100 mg/l
End point: Respiration inhibition
Exposure time: 3 h
Test Type: static test
Analytical monitoring: no
Method: OECD 209
GLP: yes

EC50 (Activated sludge): > 100 mg/l
End point: Respiration inhibition
Exposure time: 3 h
Test Type: static test
Analytical monitoring: no
Method: OECD 209
GLP: yes

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Toxicity to fish : LC50 (Rainbow trout (Salmo gairdneri)): 0,75 mg/l
Exposure time: 96 h
Test Type: semi-static test
Method: OECD 203 / ISO 7346
GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 0,23 mg/l
Exposure time: 48 h
Test Type: static test
Method: OECD Test Guideline 202
GLP: yes

Toxicity to algae/aquatic plants : ErC50 (Pseudokirchneriella subcapitata (green algae)): 0,47 mg/l
Exposure time: 72 h
Test Type: static test
Method: OECD Test Guideline 201
GLP: yes

ErC10 (Pseudokirchneriella subcapitata (green algae)): 0,119 mg/l

		Exposure time: 72 h Test Type: static test Method: OECD Test Guideline 201 GLP: yes
M-Factor (Acute aquatic toxicity)	:	1
Toxicity to microorganisms	:	EC50 (Activated sludge): > 10.000 mg/l End point: Respiration inhibition Exposure time: 3 h Method: OECD 209 / ISO 8192 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC: >= 0,0141 mg/l End point: Reproduction rate Exposure time: 21 d Species: Daphnia magna (Water flea) Test Type: semi-static test GLP: yes

Pentadecan-15-olide:

Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): > 0,17 mg/l End point: Immobilization Exposure time: 48 h Test Type: semi-static test Analytical monitoring: yes Method: OECD Test Guideline 202 GLP: yes Remarks: No effect in the area of water solubility of the substance
Toxicity to algae/aquatic plants	:	ErC10 (Desmodesmus subspicatus (green algae)): 0,421 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: Directive 67/548/EEC, Annex V, C.3. GLP: yes
Toxicity to microorganisms	:	EC50 (Activated sludge): > 10.000 mg/l End point: Respiration inhibition Exposure time: 30 min Test Type: Respiration inhibition Analytical monitoring: no Method: OECD 209 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC: 0,068 mg/l End point: Reproduction rate Exposure time: 21 d Species: Daphnia magna (Water flea)

Test Type: semi-static test
Analytical monitoring: yes
Method: OECD 211
GLP: yes

benzyl salicylate:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): 1,03 mg/l
End point: mortality
Exposure time: 96 h
Test Type: semi-static test
Analytical monitoring: yes
Method: Directive 67/548/EEC, Annex V, C.1.
GLP: yes

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna): 1,16 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 202
GLP: yes

Toxicity to algae/aquatic plants : EC50 (Pseudokirchneriella subcapitata (green algae)): 1,29 mg/l
End point: Growth rate
Exposure time: 72 h
Method: OECD Test Guideline 201
GLP: yes

NOEC (Pseudokirchneriella subcapitata (green algae)): 0,502 mg/l
End point: Growth rate
Exposure time: 72 h
Method: OECD Test Guideline 201
GLP: yes

Toxicity to soil dwelling organisms : LC50: > 1.000 mg/kg
Exposure time: 14 d
End point: mortality
Species: Eisenia fetida (earthworms)
Method: OECD Test Guideline 207
GLP:yes

cis-2-tert-Butylcyclohexyl acetate:

Toxicity to fish : LC50 (Danio rerio (zebra fish)): 1,7 mg/l
Exposure time: 96 h
Test Type: semi-static test
Method: Directive 67/548/EEC, Annex V, C.1.
GLP: yes

LC50 (Danio rerio (zebra fish)): 5,6 mg/l
Exposure time: 96 h

		Test Type: semi-static test Analytical monitoring: yes Method: Directive 67/548/EEC, Annex V, C.1. GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 17 mg/l Exposure time: 48 h Test Type: static test Method: Directive 67/548/EEC, Annex V, C.2.
Toxicity to algae/aquatic plants	:	EC50 (Desmodesmus subspicatus (green algae)): 4,2 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes NOEC (Desmodesmus subspicatus (green algae)): 0,57 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes
Toxicity to fish (Chronic toxicity)	:	EC10: 0,91 mg/l End point: mortality Exposure time: 33 d Species: Pimephales promelas (fathead minnow) Test Type: flow-through test Analytical monitoring: yes Method: OECD 210 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	0,99 mg/l End point: Reproduction rate Exposure time: 21 d Species: Daphnia magna (Water flea) Test Type: semi-static test Analytical monitoring: yes Method: OECD 211 GLP: yes

2,6-Dimethyloct-7-en-2-ol:

Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna): 38 mg/l End point: Immobilization Exposure time: 48 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 202 GLP: yes

Toxicity to algae/aquatic plants	:	EC50 (Desmodesmus subspicatus (green algae)): 80 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes NOEC (Desmodesmus subspicatus (green algae)): 25 mg/l Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes

Oxacycloheptadec-10-en-2-one:

M-Factor (Acute aquatic toxicity)	:	10
M-Factor (Chronic aquatic toxicity)	:	10

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Toxicity to fish	:	LC50 (Danio rerio (zebra fish)): > 0,999 mg/l End point: mortality Exposure time: 96 h Test Type: semi-static test Analytical monitoring: yes Method: OECD Test Guideline 203 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 0,696 mg/l End point: Immobilization Exposure time: 48 h Test Type: semi-static test Analytical monitoring: yes Method: OECD Test Guideline 202 GLP: yes
Toxicity to algae/aquatic plants	:	EC50 (Pseudokirchneriella subcapitata (green algae)): > 0,1 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes EC10 (Pseudokirchneriella subcapitata (green algae)): > 0,1 mg/l End point: Growth rate Exposure time: 72 h

		Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes
M-Factor (Acute aquatic toxicity)	:	1
Toxicity to microorganisms	:	NOEC (Activated sludge): >= 1.000 mg/l End point: Respiration inhibition Exposure time: 3 h Test Type: static test Method: OECD 209 GLP: yes
M-Factor (Chronic aquatic toxicity)	:	1
Toxicity to soil dwelling organisms	:	EC10: 248 mg/kg Exposure time: 28 d End point: Reproduction Species: Eisenia fetida (earthworms) Method: OECD Test Guideline 222 GLP:yes 72,1 mg/kg Exposure time: 28 d End point: Reproduction Species: Eisenia fetida (earthworms) Method: OECD Test Guideline 222 GLP:yes

(Ethoxymethoxy)cyclododecane:

Toxicity to fish	:	LC50 (Danio rerio (zebra fish)): 1,9 mg/l End point: mortality Exposure time: 96 h Test Type: semi-static test Analytical monitoring: yes Method: OECD Test Guideline 203 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 1,6 mg/l End point: Immobilization Exposure time: 48 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 202 GLP: yes
Toxicity to algae/aquatic plants	:	EC50 (Pseudokirchneriella subcapitata (green algae)): > 2 mg/l End point: Growth rate Exposure time: 72 h

Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

NOEC (Pseudokirchneriella subcapitata (green algae)): 0,89 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

Toxicity to microorganisms : NOEC (Activated sludge): >= 1.000 mg/l
End point: Respiration inhibition
Exposure time: 3 h
Test Type: static test
Analytical monitoring: no
Method: OECD 209
GLP: yes

(R)-p-mentha-1,8-diene; d-limonene:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): 0,720 mg/l
End point: mortality
Exposure time: 96 h
Test Type: flow-through test
Analytical monitoring: yes
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 0,307 mg/l
End point: Immobilization
Exposure time: 48 h
Test Type: semi-static test
Analytical monitoring: yes
Method: OECD Test Guideline 202
GLP: yes

Toxicity to algae/aquatic plants : EC50 (Raphidocelis subcapitata (freshwater green alga)): 0,32 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test
Analytical monitoring: yes
Method: OECD Test Guideline 201
GLP: yes

EC10 (Raphidocelis subcapitata (freshwater green alga)): 0,174 mg/l
End point: Growth rate
Exposure time: 72 h
Test Type: static test

		Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes
M-Factor (Acute aquatic toxicity)	:	1
Toxicity to microorganisms	:	EC50 (Activated sludge): 209 mg/l Exposure time: 3 h Test Type: static test Analytical monitoring: no Method: OECD Test Guideline 209 GLP: yes EC10 (Activated sludge): 18 mg/l Exposure time: 3 h Test Type: static test Analytical monitoring: no Method: OECD Test Guideline 209 GLP: yes
Toxicity to fish (Chronic toxicity)	:	EC10: > 0,37 - < 0,67 mg/l Exposure time: 8 d Species: Pimephales promelas (fathead minnow) Test Type: semi-static test Analytical monitoring: yes Method: OECD 212 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	EC10: 0,153 mg/l End point: Reproduction rate Exposure time: 21 d Species: Daphnia magna (Water flea) Test Type: semi-static test Analytical monitoring: yes Method: OECD Test Guideline 211 GLP: yes

musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone:

Toxicity to fish	:	LC50 (Poecilia reticulata (guppy)): > 0,385 mg/l End point: mortality Exposure time: 96 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 203 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): > 0,432 mg/l End point: Immobilization Exposure time: 48 h Test Type: static test

		Analytical monitoring: yes Method: OECD Test Guideline 202 GLP: yes
Toxicity to algae/aquatic plants	:	EC50 (Pseudokirchneriella subcapitata (green algae)): 0,244 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes EC10 (Pseudokirchneriella subcapitata (green algae)): 0,070 mg/l End point: Growth rate Exposure time: 72 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 201 GLP: yes
M-Factor (Acute aquatic toxicity)	:	1
Toxicity to microorganisms	:	EC50 (Activated sludge): > 1.000 mg/l Exposure time: 3 h Method: OECD 209 GLP: yes
M-Factor (Chronic aquatic toxicity)	:	1
Linalyl acetate:		
Toxicity to fish	:	LC50 (Cyprinus carpio (Carp)): 11 mg/l End point: mortality Exposure time: 96 h Test Type: flow-through test Analytical monitoring: yes Method: OECD Test Guideline 203 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 59 mg/l End point: Immobilization Exposure time: 48 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 202 GLP: yes
Toxicity to algae/aquatic plants	:	EC10 (Desmodesmus subspicatus (green algae)): 54,3 mg/l End point: Growth rate Exposure time: 96 h

		Test Type: static test Analytical monitoring: no Method: DIN 38412 (part 9) GLP: no
		EC50 (Desmodesmus subspicatus (green algae)): 156,7 mg/l End point: Growth rate Exposure time: 96 h Test Type: static test Analytical monitoring: no Method: DIN 38412 (part 9) GLP: no
Toxicity to microorganisms	:	EC20 (Activated sludge): > 1.000 mg/l End point: Respiration inhibition Exposure time: 0,5 h Test Type: static test Analytical monitoring: no Method: ISO 8192 GLP: no

p-Mentha-1,4-diene:

Toxicity to fish	:	EC50 (Danio rerio (zebra fish)): 2,792 mg/l Exposure time: 96 h Method: OECD Test Guideline 203 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 10,189 mg/l Exposure time: 48 h Test Type: static test Method: OECD Test Guideline 202 GLP: yes
Toxicity to algae/aquatic plants	:	EC50 (Scenedesmus capricornutum (fresh water algae)): > 10,82 mg/l Exposure time: 72 h Method: OECD Test Guideline 201 GLP: yes
Toxicity to microorganisms	:	EC50 (Activated sludge): > 1.000 mg/l Exposure time: 3 h Analytical monitoring: yes Method: OECD 209 GLP: yes

isoeugenol:

Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna): 7,5 mg/l End point: Immobilization Exposure time: 48 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 202

GLP: yes

12.2 Persistence and degradability

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Biodegradability : Test Type: Closed Bottle test
Result: Readily biodegradable.
Biodegradation: 68 %
Exposure time: 28 d
Method: OECD 301D
GLP: yes

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Biodegradability : Test Type: Sturm test, OECD 301-B, (CO_2):
Result: Readily biodegradable.
Biodegradation: 72 %
Exposure time: 28 d
Method: OECD 301B
GLP: yes

Pentadecan-15-olide:

Biodegradability : Test Type: Manometric Respirometry Test
Result: Readily biodegradable.
Biodegradation: 90 %
Exposure time: 28 d
Method: OECD 301F
GLP: yes

benzyl salicylate:

Biodegradability : Test Type: Manometric respiration test
Result: Readily biodegradable.
Biodegradation: 93 %
Exposure time: 28 d
Method: OECD 301F
GLP: yes

cis-2-tert-Butylcyclohexyl acetate:

Biodegradability : Test Type: Manometric respiration test
Result: Inherently biodegradable.
Biodegradation: 61 %
Exposure time: 60 d
Method: OECD 301F
GLP: yes

Methyl 2,4-dihydroxy-3,6-dimethylbenzoate:

Biodegradability : Test Type: Manometric respiration test
Inoculum: activated sludge

Result: Readily biodegradable.
Biodegradation: 63 %
Exposure time: 28 d
Method: OECD 301F
GLP: yes

2,6-Dimethyloct-7-en-2-ol:

Biodegradability : Test Type: CO_2 Evolution Test
Result: Readily biodegradable.
Biodegradation: 72 %
Exposure time: 28 d
Method: OECD 301B
GLP: yes

3-Methylcyclopentadecan-1-one:

Biodegradability : Test Type: Manometric respiration test
Result: Readily biodegradable.
Biodegradation: 80 %
Exposure time: 28 d
Method: OECD 301F
GLP: yes

Oxacycloheptadec-10-en-2-one:

Biodegradability : Test Type: MITI Test I
Result: Readily biodegradable.
Biodegradation: 93 %
Exposure time: 28 d
Method: OECD 301C
GLP: yes

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Biodegradability : Test Type: MITI Test II
Result: Partially inherently biodegradable.
Biodegradation: 25 %
Exposure time: 28 d
Method: OECD 302C
GLP: yes

Test Type: Manometric respiration test
Result: Readily biodegradable.
Biodegradation: 62 %
Exposure time: 28 d
Method: OECD 301F
GLP: yes

(Ethoxymethoxy)cyclododecane:

Biodegradability : Test Type: CO_2 Evolution Test
Result: Not readily biodegradable.
Biodegradation: < 5 %

Exposure time: 28 d
Method: OECD 301B
GLP: yes

(R)-p-mentha-1,8-diene; d-limonene:

Biodegradability : Test Type: CO2 Evolution Test
Result: Readily biodegradable.
Biodegradation: 71 %
Exposure time: 28 d
Method: OECD 301B
GLP: yes

musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone:

Biodegradability : Test Type: MITI Test II
Inoculum: activated sludge
Result: Not readily biodegradable.
Biodegradation: 0 %
Exposure time: 28 d
Method: OECD 302C
GLP: No information available.

Linalyl acetate:

Biodegradability : Test Type: Manometric respiration test
Result: Readily biodegradable.
Biodegradation: 76 %
Exposure time: 28 d
Method: OECD 301F
GLP: no

Pin-2(10)-ene:

Biodegradability : Test Type: Manometric respiration test
Result: Readily biodegradable.
Biodegradation: 81 %
Exposure time: 28 d
Method: OECD 301F
GLP: yes

p-Mentha-1,4-diene:

Biodegradability : Test Type: MITI test, (BOD/COD):
Result: Readily biodegradable.
Biodegradation: 94 %
Exposure time: 28 d
Method: OECD 301C

(E)-2-methoxy-4-(prop-1-enyl)phenol:

Biodegradability : Test Type: Manometric Respirometry Test
Result: Readily biodegradable.

		Biodegradation: 79 % Exposure time: 28 d Method: OECD 301F GLP: yes

isoeugenol:

Biodegradability	:	Test Type: Manometric respiration test Result: Readily biodegradable. Biodegradation: 79 % Exposure time: 28 d Method: OECD 301F GLP: yes

12.3 Bioaccumulative potential

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Partition coefficient: n-octanol/water	:	log Pow: 4,46 (25 °C) pH: 7,3 Method: OECD Test Guideline 123 GLP: yes

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Bioaccumulation	:	Species: Danio rerio (zebra fish) Exposure time: 174 h Bioconcentration factor (BCF): 244 Method: OECD Test Guideline 305 GLP: yes
Partition coefficient: n-octanol/water	:	log Pow: 5,628 (25 °C) pH: 6,7 Method: OECD Test Guideline 123 GLP: yes

Pentadecan-15-olide:

Partition coefficient: n-octanol/water	:	log Pow: 5,79 (25 °C) Method: OECD Test Guideline 123 GLP: yes

benzyl salicylate:

Partition coefficient: n-octanol/water	:	log Pow: 4,0 (35 °C) Method: OECD 117 GLP: yes

cis-2-tert-Butylcyclohexyl acetate:

Partition coefficient: n-octanol/water	:	log Pow: 4,8 (25 °C) pH: 7 Method: OECD 117

GLP: yes

Methyl 2,4-dihydroxy-3,6-dimethylbenzoate:

Partition coefficient: n-octanol/water : log Pow: 2,6 (20 °C)
Method: OECD 117
GLP: no

2,6-Dimethyloct-7-en-2-ol:

Partition coefficient: n-octanol/water : log Pow: 3,25 (40 °C)
pH: 7
Method: OECD 117
GLP: yes

3-Methylcyclopentadecan-1-one:

Partition coefficient: n-octanol/water : log Pow: 6,06 (25 °C)
pH: 5,65
Method: OECD Test Guideline 123
GLP: yes

Oxacycloheptadec-10-en-2-one:

Partition coefficient: n-octanol/water : log Pow: 6,7 (23 °C)
Method: OECD 117
GLP: yes

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Partition coefficient: n-octanol/water : log Pow: 5,662 (25 °C)
pH: 5,7
Method: OECD Test Guideline 123
GLP: yes

(Ethoxymethoxy)cyclododecane:

Bioaccumulation : Species: Cyprinus carpio (Carp)
Exposure time: 28 d
Bioconcentration factor (BCF): 340 - 580
Method: OECD Test Guideline 305
GLP: yes

Partition coefficient: n-octanol/water : log Pow: 5,4 (25 °C)
Method: OECD Test Guideline 123
GLP: yes

(R)-p-mentha-1,8-diene; d-limonene:

Partition coefficient: n-octanol/water : log Pow: 4,38 (37 °C)
pH: 7,2
Method: OECD Test Guideline 117

musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone:

Partition coefficient: n-octanol/water	:	log Pow: 4,24 (25 °C) Method: OECD Test Guideline 117 GLP: No information available.

Linalyl acetate:

Partition coefficient: n-octanol/water	:	log Pow: 3,9 (25 °C) Method: OECD Test Guideline 107 GLP: yes

Pin-2(10)-ene:

Partition coefficient: n-octanol/water	:	log Pow: 5,4 (35 °C)

p-Mentha-1,4-diene:

Partition coefficient: n-octanol/water	:	log Pow: 5,4 (25 °C) Method: OECD Test Guideline 117 GLP: no

(E)-2-methoxy-4-(prop-1-enyl)phenol:

Partition coefficient: n-octanol/water	:	log Pow: 2,1 (25 °C) Method: OECD 117 GLP: yes

isoeugenol:

Partition coefficient: n-octanol/water	:	log Pow: 2,55

12.4 Mobility in soil

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Distribution among environmental compartments	:	Adsorption/Soil Medium: Soil Koc: 11000, log Koc: 4 Method: OECD 121

Pentadecan-15-olide:

Distribution among environmental compartments	:	log Koc: 4,65 Method: OECD 121

benzyl salicylate:

Distribution among environmental compartments	:	Adsorption/Soil Medium: Soil log Koc: 3,75 Method: OECD 121

2,6-Dimethyloct-7-en-2-ol:

Distribution among environmental compartments : Medium: Soil
Koc: 177,83, log Koc: 2,25
Method: OECD 121

Oxacycloheptadec-10-en-2-one:

Distribution among environmental compartments : Adsorption/Soil
Medium: Soil
Koc: 2,209
Method: OECD 121

(Ethoxymethoxy)cyclododecane:

Distribution among environmental compartments : Adsorption/Soil
log Koc: 4,165
Remarks: calculated

12.5 Results of PBT and vPvB assessment

Product:

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate**:**

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Pentadecan-15-olide:

Assessment : Substance is not persistent, bioaccumulative, and toxic (PBT).. Substance is not very persistent and very bioaccumulative (vPvB).

(Ethoxymethoxy)cyclododecane:

Assessment : Substance is not persistent, bioaccumulative, and toxic (PBT).. Substance is not very persistent and very bioaccumulative (vPvB).

12.6 Endocrine disrupting properties

Product:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation

(EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Pentadecan-15-olide:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

12.7 Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

Components:

(1S, 1'R)-[1-(3', 3'-dimethyl-1'-cyclohexyl)ethoxycarbonyl]methyl propanoate:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

reaction mass of cis-and trans-cyclohexadec-8-en-1-one:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

Pentadecan-15-olide:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

Methyl 2,4-dihydroxy-3,6-dimethylbenzoate:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

2,6-Dimethyloct-7-en-2-ol:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

3-Methylcyclopentadecan-1-one:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

Oxacycloheptadec-10-en-2-one:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

2,2,6-Trimethyl-α-propylcyclohexanepropanol:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

(Ethoxymethoxy)cyclododecane:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.
Empty containers should be taken to an approved waste handling site for recycling or disposal.
Do not re-use empty containers.

SECTION 14: Transport information

14.1 UN number or ID number

ADR : UN 3082

RID : UN 3082

IMDG : UN 3082

IATA : UN 3082

14.2 UN proper shipping name

ADR : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Oxacycloheptadec-10-en-2-one)

RID : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID,

		N.O.S. (Oxacycloheptadec-10-en-2-one)
IMDG	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Oxacycloheptadec-10-en-2-one, CIS, TRANS-CYCLOHEXADEC-8-EN-1-ONE)
IATA	:	Environmentally hazardous substance, liquid, n.o.s. (Oxacycloheptadec-10-en-2-one)

14.3 Transport hazard class(es)

		Class	Subsidiary risks
ADR	:	9	
RID	:	9	
IMDG	:	9	
IATA	:	9	

14.4 Packing group

ADR

Packing group	:	III
Classification Code	:	M6
Hazard Identification Number	:	90
Labels	:	9
Tunnel restriction code	:	(-)

RID

Packing group	:	III
Classification Code	:	M6
Hazard Identification Number	:	90
Labels	:	9

IMDG

Packing group	:	III
Labels	:	9
EmS Code	:	F-A, S-F

IATA (Cargo)

Packing instruction (cargo aircraft)	:	964
Packing instruction (LQ)	:	Y964
Packing group	:	III
Labels	:	Miscellaneous

IATA (Passenger)

Packing instruction (passen-ger aircraft)	:	964
Packing instruction (LQ)	:	Y964
Packing group	:	III
Labels	:	Miscellaneous

14.5 Environmental hazards

ADR
Environmentally hazardous : yes

RID
Environmentally hazardous : yes

IMDG
Marine pollutant : yes

14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

14.7 Maritime transport in bulk according to IMO instruments

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

REACH - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles (Annex XVII) : Conditions of restriction for the following entries should be considered: Number on list 3: 5-Isopropyl-2-methylbicyclo[3.1.0]hex-2-ene, Terpineol, acetate, p-Mentha-1,4-diene, (1S)-3,7,7-Trimethylbicyclo[4.1.0]hept-3-ene, Isopropyl hexanoate, (E)-4-(2,6,6-Trimethyl-1-cyclohexen-1-yl)-3-buten-2-one, 2-tert-Butylcyclohexyl acetate, Thuj-4(10)-ene, p-Menth-1-en-8-ol, 7-Methyl-3-methyleneocta-1,6-diene, Caryophyllene, 3,7-Dimethylocta-1,3,6-triene, Nerol, Geranyl acetate, p-Mentha-1(7),2-diene, Oct-1-en-3-yl acetate, 3-Methyl-4-(2,6,6-trimethyl-1-cyclohexen-1-yl)-3-buten-2-one, 2,6,10-Trimethyldodeca-2,6,9,11-tetraene, (R*,R*)-α,4-Dimethyl-α-(4-methyl-3-pentenyl)cyclohex-3-ene-1-methanol, 1H-Indene-ar-propanal, 2,3-dihydro-1,1-dimethyl-, cis-2-tert-Butylcyclohexyl acetate, 1,6-Cyclodecadiene, 1-methyl-5-methylene-8-(1-methylethyl), Neryl acetate, METHOXYAMBROCENIDE, 2,6-Dimethyloct-7-en-2-ol, Ethyl 2-methylvalerate, Dimethylcyclohex-3-

ene-1-carbaldehyde, 4-(2,5,6,6-Tetramethylcyclohex-2-enyl)but-3-en-2-one, cis-Hex-3-en-1-ol, 3-Methylcyclopentadecan-1-one, Oxacycloheptadec-10-en-2-one, 2,2,6-Trimethyl-α-propylcyclohexanepropanol, (Ethoxymethoxy)cyclododecane, Allyl heptanoate, 4-Allylanisole, methanol, trans-Hex-2-enal, Eugenol, Linalyl acetate, Cineole, Farnesol, 3-Methyl-4-(2,6,6-trimethyl-2-cyclohexen-1-yl)-3-buten-2-one, 4-tert-Butylcyclohexyl acetate, trans-Hex-2-en-1-ol, p-Mentha-1,4(8)-diene, Pin-2(10)-ene, Pin-2(3)-ene, p-Menth-1-en-4-ol

Number on list 3

Number on list 28: safrole; 5-allyl-1,3-benzodioxole

Number on list 40: Ethyl 2-methylvalerate, cis-Hex-3-en-1-ol, methanol, trans-Hex-2-enal, Cineole, Bornan-2-one, hexan-1-ol, trans-Hex-2-en-1-ol, Pin-2(10)-ene, Pin-2(3)-ene, 5-Isopropyl-2-methylbicyclo[3.1.0]hex-2-ene, p-Mentha-1,4-diene, (1S)-3,7,7-Trimethylbicyclo[4.1.0]hept-3-ene, Isopropyl hexanoate, Thuj-4(10)-ene, p-mentha-1,3-diene; 1-isopropyl-4-methylcyclohexa-1,3-diene; alpha-terpinene, 7-Methyl-3-methyleneocta-1,6-diene, 3,7-Dimethylocta-1,3,6-triene, p-Mentha-1(7),2-diene, DL-borneol

Number on list 75: If you intend to use this product as tattoo ink, please contact your vendor.

Number on list 200-659-6: methanol

REACH - Candidate List of Substances of Very High Concern for Authorisation (Article 59).	:	Not applicable
REACH - List of substances subject to authorisation (Annex XIV)	:	Not applicable

Seveso III: Directive 2012/18/EU of the European Parliament and of the Council on the control of major-accident hazards involving dangerous substances.

		Quantity 1	Quantity 2
E1	ENVIRONMENTAL HAZARDS	100 t	200 t
	E2		

Water hazard class (Germany) : WGK 2 obviously hazardous to water
Classification according to AwSV, Annex 1 (5.2)

TA Luft List (Germany) : 5.2.1: Total dust:
others: 3,19 % Methyl 2,4-dihydroxy-3,6-dimethylbenzoate, [1R-(1R*,4R*,6R*,10S*)]-4,12,12-Trimethyl-9-methylene-5-oxatricyclo[8.2.0.04,6]dodecane, Pentaerythritol tetrakis(3-(3,5-di-tert-butyl-4-hydroxyphenyl)propionate)
5.2.5: Organic Substances:
Class 1: 19,08 % (2R,4R)-4-methyl-2-(2-methylprop-1-enyl)oxane, (R)-5-Isopropyl-2-methylcyclohexa-1,3-diene, 3,7,11-Trimethyldodeca-1,6,10-trien-3-ol,mixed isomers, Methyl benzoate, 4-Methylanisole, (S)-p-mentha-1,8-diene; l-limonene, (1S)-3,7,7-Trimethylbicyclo[4.1.0]hept-3-ene, p-Mentha-1,4-diene, 1-isopropyl-4-methylbenzene; p-cymene, Pin-2(3)-ene, Pin-2(10)-ene, p-Mentha-1,4(8)-diene, Farnesol, Allyl hexanoate, trans-Hex-2-enal, benzyl alcohol, 2-phenoxyethanol, 4-Allylveratrole, 4-Allylanisole, Allyl heptanoate, reaction mass of cis-and trans-cyclohexadec-8-en-1-one, 2,2,6-Trimethyl-α-propylcyclohexanepropanol, Oxacycloheptadec-10-en-2-one, 3-Methylcyclopentadecan-1-one, musk ketone; 3,5-dinitro-2,6-dimethyl-4-tert-butylacetophenone; 4'-tert-butyl-2', 6'-dimethyl-3',; 5'-dinitroacetophenone, musk xylene; 5-tert-butyl-2,4,6-trinitro-m-xylene, Bornan-2-one, methanol, safrole; 5-allyl-1,3-benzodioxole
5.2.7.1.1: Carcinogenic substance:
Class 1: < 0,01 % safrole; 5-allyl-1,3-benzodioxole

Volatile organic compounds : Directive 2010/75/EU of 24 November 2010 on industrial and livestock rearing emissions (integrated pollution prevention and control)
Volatile organic compounds (VOC) content: 3,98 %

Other regulations:

Take note of Law on the protection of mothers at work, in education and in studies (Maternity Protection Act - MuSchG).
Take note of Directive 94/33/EC on the protection of young people at work or stricter national regulations, where applicable.

15.2 Chemical safety assessment

No Chemical Safety Assessment has been carried out for this mixture.

SECTION 16: Other information

Full text of H-Statements

H226	:	Flammable liquid and vapour.
H302	:	Harmful if swallowed.
H304	:	May be fatal if swallowed and enters airways.
H312	:	Harmful in contact with skin.
H315	:	Causes skin irritation.
H317	:	May cause an allergic skin reaction.
H319	:	Causes serious eye irritation.
H332	:	Harmful if inhaled.
H335	:	May cause respiratory irritation.
H336	:	May cause drowsiness or dizziness.
H351	:	Suspected of causing cancer.
H361	:	Suspected of damaging fertility or the unborn child.
H400	:	Very toxic to aquatic life.
H410	:	Very toxic to aquatic life with long lasting effects.
H411	:	Toxic to aquatic life with long lasting effects.
H412	:	Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	:	Acute toxicity
Aquatic Acute	:	Short-term (acute) aquatic hazard
Aquatic Chronic	:	Long-term (chronic) aquatic hazard
Asp. Tox.	:	Aspiration hazard
Carc.	:	Carcinogenicity
Eye Irrit.	:	Eye irritation
Flam. Liq.	:	Flammable liquids
Repr.	:	Reproductive toxicity
Skin Irrit.	:	Skin irritation
Skin Sens.	:	Skin sensitisation
STOT SE	:	Specific target organ toxicity - single exposure
DE TRGS 900	:	Germany. TRGS 900 - Occupational exposure limit values.
DFG	:	Senate commission for the review of compounds at the work place dangerous for the health (MAK-commission).
DE TRGS 900 /	:	Exposure limit(s)
DFG / MAK	:	Maximum allowable concentration:

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China;

IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

Classification of the mixture:		**Classification procedure:**
Skin Sens. 1	H317	Calculation method
Aquatic Acute 1	H400	Calculation method
Aquatic Chronic 1	H410	Calculation method

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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